CYBERSECURITY / ETHICAL HACKING

Learning Roadmap

Zero theke junior job, tarpor international remote e senior level — 15 ta module, ekta por ekta.

15	18–24	3–5	2–3
module	mash · junior job-ready	bochor · senior remote	ghonta · protidin

Module order

Upor theke niche, ekta shesh kore porer ta. Shomoy zero theke, din e 2–3 ghonta dhore.

PART A · JUNIOR JOB-READY (MODULE 1–11)

01	**Computer Basics**	Mash 1–3
02	**Networking**	Mash 4–5
03	**Linux + Windows**	Mash 5–6
04	**Programming — Python + Bash**	Mash 6–7
05	**Web Basics + Tools + Lab**	Mash 7
06	**Web Application Hacking**	Mash 8–10
07	**Network Pentest + Privilege Escalation**	Mash 10–11
08	**Report Writing + First Cert**	Mash 11–13
09	**AI + AI Security**	Mash 12–13
10	**Specialization + Real Bugs**	Mash 13–18
11	**Portfolio + Job Hunt**	Mash 16–24

PART B · ADVANCED REMOTE (MODULE 12–15)

12	**Deep Specialization + Advanced Cert**	Mash 24–30
13	**Senior-level Core Skills**	Mash 24–36
14	**Remote Work Skills**	Mash 24–36
15	**Paid Experience → Senior Remote Job**	Mash 24–60

3 ta rule shob module e:

1. Protiti module e **output** rakho — notes, writeup, GitHub. Shudhu video dekha shekha na.
2. **Consistency** — protidin 2 ghonta, weekend e 12 ghonta na.
3. **Permission chara** kono system e test kora illegal — shudhu nijer lab, practice site, ba written permission.

MODULE 1

Computer Basics

Mash 1–3 · Computer already jana thakle skip

Goal: Computer ke nijer haater moto chalano, ar online e nijeke safe rakha.

KI SHIKHBE

1 **Computer chalano**
Hardware (CPU, RAM, SSD), shut down / restart / sleep, mouse, keyboard shortcut

2 **Typing**
Din e 20–30 min practice. Target 30+ WPM, pore 40+

3 **Windows**
File/folder, file extension, software install (official site theke), Task Manager, settings

4 **Internet + Google search**
English keyword, “quotes”, error message copy kore search — hacker der #1 skill

5 **Online safety**
Password manager (Bitwarden), 2FA, phishing chena, bKash/Nagad OTP scam

6 **Number system + logic**
Binary, decimal, hex conversion; AND / OR / NOT

7 **Command line intro**
CMD/PowerShell: cd, dir, mkdir, ipconfig, ping

8 **Virtual machine**
VirtualBox install kore Ubuntu chalano

9 **Technical English**
Din e 20–30 min English tech video, din e 5 ta notun word

KOTHAY PRACTICE

keybr.com, typing.com, GCFGlobal Computer Basics, Professor Messer (CompTIA A+), TryHackMe Pre-Security

MODULE SHESH E TUMI PARBE

- ☐ Kono help chara software install, file organize korte
- ☐ Error Google kore nije solve korte
- ☐ VirtualBox e Ubuntu chalate
- ☐ Subtitle soho English tech video bujhte

MODULE 2

Networking

Mash 4–5

Goal: Internet e data kivabe jay sheta bujha — hacking er bhit eta.

KI SHIKHBE

1. **OSI ar TCP/IP model**
   Protiti layer e ki hoy
2. **TCP vs UDP**
   3-way handshake
3. **IP address + subnetting**
   Network ar host bhag kora
4. **DNS, DHCP**
   Domain theke IP, IP kivabe pay
5. **HTTP / HTTPS + TLS basics**
   Website kivabe load hoy
6. **Common ports**
   21, 22, 25, 53, 80, 443, 445, 3306, 3389
7. **Wireshark**
   Packet capture kore traffic dekha

KOTHAY PRACTICE

TryHackMe Pre-Security, Professor Messer Network+ (free video)

MODULE SHESH E TUMI PARBE

- ☐ Browser e URL dile ki ki hoy step by step bujhate
- ☐ Subnet calculate korte
- ☐ Wireshark e HTTP request khuje ber korte

MODULE
3

Linux + Windows

Mash 5–6

Goal: Linux terminal comfortably chalano ar Windows er security basic bujha.

KI SHIKHBE

1. **Linux file system**
   Directory structure, file khuja
2. **Permissions**
   chmod, chown, SUID, users ar groups
3. **Process + cron**
   Running process dekha, scheduled task
4. **Text tools**
   grep, awk, sed, pipe
5. **ssh + package management**
   Remote login, software install (apt)
6. **Windows security basic**
   Active Directory concept (user, group, domain), PowerShell, services, registry

KOTHAY PRACTICE

OverTheWire Bandit, TryHackMe Linux Fundamentals

MODULE SHESH E TUMI PARBE

- ☐ Terminal diye file khuje, permission bodlate
- ☐ Bandit er beshirbhag level solve korte
- ☐ AD ki sheta bujhate

MODULE
4

Programming — Python + Bash

Mash 6–7

Goal: Nijer kaj automate korar moto code likhte para.

KI SHIKHBE

1 **Python basics**
Variable, if/else, loop, function, list/dict, file

2 **Python for security**
requests (web), sockets (network), choto automation script

3 **Bash scripting**
Linux e repetitive kaj automate

KOTHAY PRACTICE

Python for Everybody, CS50P (free)

MODULE SHESH E TUMI PARBE

- ☐ Ekta website e request pathanor script likhte
- ☐ Ekta simple port scanner likhte
- ☐ Script GitHub e rakhte

MODULE
5

Web Basics + Tools + Lab

Mash 7

Goal: Website kivabe kaj kore bujha, ar nijer hacking lab banano.

KI SHIKHBE

1 **HTTP in detail**
Request/response, headers, cookies, sessions

2 **Web security concept**
CORS, same-origin policy

3 **API**
REST API, JSON, auth (session vs JWT)

4 **Tools**
Burp Suite (Community), Nmap, curl, browser DevTools

5 **Home lab**
VirtualBox + Kali Linux + Metasploitable / DVWA

KOTHAY PRACTICE

TryHackMe Cyber Security 101 path, PortSwigger Academy (shuru)

MODULE SHESH E TUMI PARBE

- ☐ Burp diye request dhore modify korte
- ☐ Nmap diye scan korte
- ☐ Nijer lab chalu kore ekta README/blog likhte

MODULE
6

Web Application Hacking

Mash 8–10 · shobcheye important module

Goal: Real website er common bug khuje exploit korte para.

KI SHIKHBE

1. **OWASP Top 10**
   Shob gula ki, keno hoy, kivabe fix
2. **SQL Injection**
   In-band, blind, time-based; manual + sqlmap
3. **XSS**
   Reflected, stored, DOM
4. **Broken Access Control / IDOR**
   Shobcheye common real-world bug — beshi shomoy dao
5. **Authentication flaws**
   Password reset logic, 2FA bypass
6. **CSRF, SSRF**
7. **File upload, Path traversal, LFI**
8. **Business logic flaws**
   AI eta kom pare — tomar edge
9. **API security**
   OWASP API Top 10

KOTHAY PRACTICE

PortSwigger Web Security Academy (free, best) — target 100+ lab

MODULE SHESH E TUMI PARBE

- ☐ OWASP Top 10 er protiti ta exploit + fix explain korte
- ☐ 100+ PortSwigger lab complete
- ☐ Protiti lab er notes

MODULE 7

Network Pentest + Privilege Escalation

Mash 10–11

Goal: Ekta machine e dhuke full control neoa porjonto shikha.

KI SHIKHBE

1. **Recon + enumeration**
   Nmap, service khuje ber kora
2. **Service exploitation**
   Vulnerable service theke access
3. **Privilege escalation**
   Linux ar Windows e admin/root howa
4. **Password attack basics**
   Hash, brute force concept
5. **Active Directory intro**
   Kerberoasting, BloodHound concept

KOTHAY PRACTICE

TryHackMe Jr Penetration Tester path, HackTheBox Starting Point + Easy machine

MODULE SHESH E TUMI PARBE

- ☐ 20–30 ta THM/HTB machine solve
- ☐ Protiti machine er notes (Obsidian/Notion)

MODULE
8

Report Writing + First Cert

Mash 11–13

Goal: Kaj ke professional bhabe dekhano, ar prothom hands-on certificate.

KI SHIKHBE

1. **Pentest report**
   Executive summary, finding, severity (CVSS), proof, fix
2. **Writeup**
   Retired HTB machine ba PortSwigger lab er writeup blog/GitHub e
3. **eJPT certificate**
   Prothom hands-on cert hisebe best

KOTHAY PRACTICE

Nijer lab (DVWA) er upor full report banao; INE eJPT

MODULE SHESH E TUMI PARBE

- [ ] 1 ta professional sample report
- [ ] 3–5 ta public writeup
- [ ] eJPT pass

MODULE
9

AI + AI Security

Mash 12–13

Goal: AI ke tool hisebe use kora, ar AI system er bug khuja.

KI SHIKHBE

1 **AI as a tool**
Recon/log/code review druto korte, script likhte — kintu protiti finding nije verify

2 **OWASP Top 10 for LLM**
AI app er common risk

3 **Prompt injection**
Direct + indirect, data leak, agent/tool abuse

KOTHAY PRACTICE

Gandalf (Lakera), PortSwigger LLM attack labs, nijer choto chatbot banie test

MODULE SHESH E TUMI PARBE

- ☐ Prompt injection ki, kivabe hoy explain korte
- ☐ AI diye kaj korleo manually explain korte

MODULE
10

Specialization + Real Bugs

Mash 13–18

Goal: Ekta lane bechhe nao ar real-world e bug khujo.

KI SHIKHBE

1 **Ekta lane bechho**
Web AppSec / WordPress security / Cloud / AI security / SOC (blue team)

2 **Bug bounty**
HackerOne, Bugcrowd, Wordfence, Patchstack — scope ar rules line-by-line porho

3 **Responsible disclosure**
Vendor ke report, fix er pore publish

4 **Legal rules**
Permission chara kono system e test na; real user data download na; DoS na

KOTHAY PRACTICE

Bug bounty platform, open-source code audit, Semgrep

MODULE SHESH E TUMI PARBE

- ☐ 1–3 ta real finding (valid bounty / CVE / disclosure)
- ☐ Nijer lane er 1–2 ta GitHub tool/script

MODULE 11

Portfolio + Job Hunt

Mash 16–24

Goal: Prothom job/client paoa.

KI SHIKHBE

1. **Portfolio**
   GitHub (tool), blog (writeup), sample report, THM/HTB profile
2. **Resume**
   Summary → Skills → Projects & Findings → Certs → Experience → Links
3. **LinkedIn**
   Weekly learning post, writeup share
4. **Apply**
   Mass apply na — protiti application e best project/finding dekhao
5. **Interview prep**
   OWASP Top 10 deep explain, ekta bug live walk-through
6. **OSCP (optional)**
   Remote/international job e beshi chay

KOTHAY PRACTICE

LinkedIn, company career page, Upwork (WordPress security audit), community referral

MODULE SHESH E TUMI PARBE

- ☐ Updated resume + LinkedIn + portfolio link
- ☐ 70% ready holei apply shuru

MODULE
12

Deep Specialization + Advanced Cert

Mash 24–30 · Advanced remote shuru

Goal: Ekta track e senior level er gobhir skill. Remote e shera combo: Web AppSec + Cloud.

KI SHIKHBE

1 **Advanced Web / AppSec**
Request smuggling, cache poisoning, race condition, OAuth/SAML/JWT attack, source code review → BSCP, OSWE

2 **Cloud / DevSecOps**
AWS IAM deep, Kubernetes security, Terraform, CI/CD security → AWS Security Specialty, CKS

3 **Red Team**
Active Directory deep, ADCS, Entra ID, C2 (nijer lab e) → CRTO, OSEP

4 **AI / LLM Security**
Agent/tool abuse, RAG attack, model supply chain; garak, PyRIT

5 **Detection Engineering**
Sigma, KQL/SPL, EDR telemetry, incident response

6 **Smart Contract (Web3)**
Solidity bug, Foundry; Code4rena/Sherlock competitive audit

KOTHAY PRACTICE

Ekta primary + ekta secondary track bechhe nao. Practice: PortSwigger Expert labs, flaws.cloud, CloudGoat, GOAD lab

MODULE SHESH E TUMI PARBE

- ☐ Ekta track e advanced cert
- ☐ 3–5 ta deep research blog post

MODULE
13

Senior-level Core Skills

Mash 24–36

Goal: Junior theke senior er parthokko: design bujha, risk bujhano, team ke guide kora.

KI SHIKHBE

1. **Code review**
   2–3 language e (Java, Node/TS, Python, PHP, Go) source theke sink trace
2. **Threat modeling**
   Data flow diagram, STRIDE — design e risk dhora
3. **Risk communication**
   CVSS v4 + business impact; CTO ke 5 minute e bujhano
4. **Cloud (AWS) hands-on**
   Nijer track e apply korte para
5. **Compliance basics**
   SOC 2, ISO 27001, PCI DSS — client keno test korache
6. **Automation + AI**
   LLM API diye nijer security workflow automate

KOTHAY PRACTICE

Open-source project e code review, nijer project er threat model likho

MODULE SHESH E TUMI PARBE

- ☐ Ekta app er threat model banate
- ☐ Non-technical manush ke bug er impact bujhate

MODULE
14

Remote Work Skills

Mash 24–36 · Module 12–13 er sathe

Goal: International team/client er sathe kaj korar moto ready howa.

KI SHIKHBE

1 **Spoken English**
30–45 min technical interview fluently; nijer kaj English e bole record kore practice

2 **Async communication**
Clear written update, Slack, Jira, Git PR workflow

3 **Time zone**
US/EU er sathe din e 3–4 ghonta overlap

4 **Setup**
IPS/UPS, backup internet, bhalo headset + webcam

5 **Payment + contract**
Payoneer / bank wire; Deel, Remote.com er moto platform; NDA ar contract pora

KOTHAY PRACTICE

Shoptahe 1 ta mock interview, nijer best finding er 5-min English video

MODULE SHESH E TUMI PARBE

- ☐ English e confidently interview dite
- ☐ Load shedding/internet down holeo kaj chalate

MODULE
15

Paid Experience → Senior Remote Job

Mash 24–60

Goal: Remote theke-i experience banano ar senior role paoa.

KI SHIKHBE

1 **Step 1 · Freelance**
Upwork, WordPress audit, startup pentest

2 **Step 2 · Pentest platform**
Synack Red Team, Cobalt Core, HackerOne pentest (Bangladesh allowed kina check koro)

3 **Step 3 · Private bounty**
Valid report diye private program invite

4 **Step 4 · Remote contract**
EU/UK/AU er pentest firm — LinkedIn e direct message

5 **Step 5 · Full-time remote**
“Remote — Worldwide” job; US citizen/clearance lage emon job skip

6 **Reputation**
Original research blog, CVE, open-source tool, BSides talk

7 **Interview + negotiation**
Take-home report e beshi focus; BD local rate diye nijeke value koro na

KOTHAY PRACTICE

LinkedIn (Remote + Worldwide filter), Wellfound, We Work Remotely, Remote OK

MODULE SHESH E TUMI PARBE

- ☐ 12–18+ mash paid security experience
- ☐ 3+ high/critical finding
- ☐ Senior remote offer